



Wasti Murad

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Portfolio: muradohi.github.io/wasti-murad/

SUMMARY

MSc Data Science graduate with hands-on experience in end-to-end machine learning systems, predictive modeling, and big data applications. Skilled in MLOps, interpretable AI, reproducible experimentation, and data-driven storytelling, built through applied ML projects alongside ML research at Maastrro’s Clinical Data Science department and an ML internship at Maastricht UMC. A methodical problem-solver who applies the same critical thinking and strategic reasoning to data challenges as to planning a few moves ahead in chess.

EDUCATION

Maastricht University, Limburg, Maastricht, Netherlands	Sep 2023 – Aug 2025
MSc Data Science for Decision Making	GPA 3.7/4.0
Jahangirnagar University, Savar, Dhaka, Bangladesh	Mar 2017 – Sep 2022
BSc Physics	GPA 3.2/4.0

EXPERIENCE

Master’s Thesis – Artificial Intelligence Guided Adaptive Radiotherapy

Maastricht University and Maastrro (Clinical Data Science)	Maastricht, Netherlands	Feb 2025 – Aug 2025
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- Built a 3D Variational Autoencoder for anatomy-preserving counterfactuals, addressing interpretability gaps in radiotherapy AI and improving reconstruction by 4%
- Restructured the latent space to disentangle tumor and organ variations, enabling clinician-validated traversals
- Standardized experiment tracking to improve reproducibility and cross-team comparisons

Tools & Methods: Python, PyTorch, CUDA, MONAI, MLflow

Machine Learning Intern – Model Generalization and Data Pipelines

Maastricht University Medical Center	Maastricht, Netherlands	Sep 2024 – Feb 2025
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- Enhanced preprocessing and class-balancing pipelines to stabilize training on imbalanced medical datasets, improving model reliability and generalization on unseen patient data
- Unified scattered configurations and manual logging into a tracking framework for reproducible team experiments
- Produced Explainable AI summaries, linking technical outputs and clinical interpretation for stakeholder-ready insights

Tools & Methods: Python, Pandas, AutoML, Hydra, MLflow, Git, GitHub

TECHNICAL PROJECTS

Code Repository

Causal Inference and Uplift Modeling for Marketing Campaign Optimization

- Estimated individual treatment effects in observational data, correcting for selection bias for true incremental lift
- Identified a 25–30% high-uplift segment concentrating most of the campaign’s incremental revenue potential
- Designed a Target / Don’t-target / Exclude strategy, shifting allocation from broad targeting to profit-driven spend

Tools & Methods: Python, Pandas, Scikit-learn, Statsmodels, CausalML, Propensity Scoring, T-Learner, Uplift Random Forest

Predictive Maintenance and Remaining Useful Life (RUL) Prediction on NASA CMAPSS Turbofan Data

- Built an RUL prediction pipeline, engineering degradation and trend features to capture engine wear patterns
- Developed an interactive dashboard with per-engine diagnostics, RUL forecasts, and risk-based maintenance triage
- Containerized the pipeline with experiment tracking to enable reproducible training and streamlined deployment

Tools & Methods: Python, PyTorch, LSTM, XGBoost, Pandas, Plotly Dash, Docker, MLflow

Apple Retail Sales and Warranty Analytics

- Surfaced revenue concentration showing the top 20% of stores drive 40% of revenue, guiding store prioritization
- Analyzed 180-day warranty cohorts, finding mid-tier products had 35% fewer claims than premium tiers
- Delivered Tableau dashboards to track regional trends and support data-driven store and product decisions

Tools & Methods: SQL, Window Functions, CTEs, Joins, Tableau

LLM-Powered E-Commerce Analytics Assistant

- Developed a hybrid query-routing system to eliminate hallucinated metrics across sales and review data
- Implemented a faithfulness and relevance evaluation framework, accelerating iteration and improving LLM reliability
- Created an interactive interface for non-technical users to query sales and reviews via natural language

Tools & Methods: Python, LangChain, OpenAI, RAG, Pandas, Streamlit, YAML

End-to-End Machine Learning Pipeline with Automated Training and Deployment

- Streamlined training, testing, and image publishing via CI/CD, replacing manual workflows with reproducible releases
- Added unit and integration tests to catch regressions early and ensure production-ready builds
- Versioned model artifacts and container images to enable traceable releases and safe rollbacks

Tools & Methods: Docker, scikit-learn, GitHub Actions, CI/CD, MLOps, pytest

Databricks Delta Live Tables: Streaming ELT Medallion Pipeline

- Engineered a streaming ELT pipeline with schema validation and quality enforcement, replacing brittle batch workflows
- Implemented transformation logic for joins, derived metrics, and type casting to produce analytics-ready datasets
- Versioned the full pipeline in Git and Databricks, enabling reproducible and peer-reviewable data engineering workflows

Tools & Methods: Databricks, PySpark, Delta Live Tables, Delta Lake, Medallion Architecture, SQL

RESEARCH EXPERIENCE

Wildlife Monitoring (Group Research Project)

Maastricht University Maastricht, Netherlands Aug 2023 – Jul 2024

- Architected an edge-to-cloud pipeline with motion and audio triggers, eliminating manual stream screening
- Fused vision, language, and audio models into a unified detection system, drastically reducing identification time
- Deployed the system behind a containerized API, enabling continuous real-time monitoring over post-hoc review

Tools & Methods: Python, OpenCV, Librosa, Transformers, FastAPI, Docker

SKILLS

Programming and Tools: Python, SQL, Excel, Git, GitHub, Tableau
Machine Learning and AI: LangChain, RAG, LLMs, PyTorch, TensorFlow, Scikit-Learn, Computer Vision, Transfer Learning, Time Series, Predictive Modeling
Data Science Libraries: Pandas, NumPy, SciPy, Statsmodels, OpenCV
Big Data and Engineering: Apache Spark, PySpark, Databricks, Delta Lake, Delta Live Tables, Medallion Architecture
Cloud and MLOps: AWS, Azure Data Factory, SageMaker, MLflow, Hydra, Docker, GitHub Actions, CI/CD
Experimentation and Analytics: A/B Testing, Statistical Testing, Power Analysis, Sequential Testing
Languages: Bengali (Native), English (C1), Dutch (A1)

AWARDS

International University Physics Competition 2019, 2020

- Bronze Medal (2020): Problem A – Protecting Travelers to Mars
- Silver Medal (2019): Problem A – Ion Thrusters to Saturn